

GASSIM, Saudi ArabiaWMO: **404050**Lat: **26.300N**Lon: **43.767E**Elev: **2126**StdP: **13.60**Time Zone: **3.00 (E03)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	39.1	42.6	N/A	N/A	N/A	N/A	N/A	N/A	21.2	60.5	18.8	62.0	2.9	30	0.476

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	29.0	113.3	N/A	111.6	N/A	110.8	N/A	N/A	N/A	N/A	N/A	N/A	N/A	8.9	30

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4)	20.3	17.9	15.7	DB	32.9	115.9	3.5	1.6	30.4	117.0	28.4	118.0	26.4	118.9	23.9	120.0
(5)				WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	78.6	57.0	61.5	68.9	78.4	88.6	93.8	95.7	96.4	91.5	82.0	68.5	59.7	
	DBStd	15.20	6.20	6.93	6.79	6.00	5.09	3.22	3.48	3.28	3.91	4.55	6.52	6.09	
	HDD50	20	12	3	0	0	0	0	0	0	0	0	0	5	
	HDD65	656	256	136	38	1	0	0	0	0	0	0	40	185	
	CDD50	10455	230	326	587	853	1196	1314	1416	1440	1244	991	554	304	
	CDD65	5616	9	39	160	404	731	864	951	975	794	526	144	19	
	CDH74	89593	119	459	1783	4888	11325	15081	17086	17408	13004	6873	1374	193	
	CDH80	61140	15	119	730	2612	7477	10893	12694	12997	9044	4031	505	23	
Wind	WSAvg	7.0	6.4	7.4	8.1	8.4	8.1	7.3	6.9	6.4	5.9	5.9	6.7	6.2	
Precipitation	PrecAvg	4.2	0.5	0.5	0.7	0.7	0.2	0.0	0.0	0.0	0.0	0.2	0.9	0.5	
	PrecMax	9.7	3.7	2.2	3.1	1.9	1.6	0.4	0.0	0.0	0.2	1.3	4.4	4.1	
	PrecMin	0.9	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	
	PrecStd	2.2	0.8	0.6	0.9	0.6	0.4	0.1	0.0	0.0	0.0	0.3	1.1	0.9	
	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	81.0	88.1	96.9	102.7	110.3	113.3	116.2	115.2	112.7	103.7	92.8	82.7
MCWB			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
2%		DB	77.2	83.8	91.7	100.0	107.7	111.4	113.3	113.4	109.7	100.8	89.5	79.0	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
5%		DB	73.5	79.9	87.9	96.5	105.5	109.6	111.5	112.6	107.9	98.8	86.1	75.4	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
10%		DB	69.7	75.5	83.9	93.0	102.4	107.8	109.7	110.9	105.8	96.6	82.4	71.8	
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures		0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
	Mean Daily Temperature Range	5% DB	MDBR	21.8	23.9	24.8	25.3	26.2	28.4	28.8	29.0	29.1	27.9	22.6	21.7
			MCDBR	26.9	28.6	30.3	28.9	28.3	29.6	29.2	29.7	30.2	30.0	27.9	26.4
MCWBR			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
5% WB		MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
Clear-Sky Solar Irradiance	taub	0.417	0.460	0.535	0.596	0.626	0.566	0.537	0.566	0.530	0.489	0.488	0.434		
	taud	1.999	1.878	1.705	1.619	1.566	1.667	1.754	1.693	1.747	1.826	1.835	1.954		
	Ebn at Noon	257	255	244	233	226	238	245	237	241	243	231	244		
	Edh at Noon	49	60	75	84	88	79	72	77	71	62	57	49		
All-Sky Solar Radiation	RadAvg	1317	1600	1902	2074	2311	2556	2480	2319	2084	1763	1335	1217		
	RadStd	88	84	81	99	105	75	63	56	60	77	110	76		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+1.46	N/A	N/A	+1.39	N/A	N/A	N/A	-103	+587	+490
Regional (0 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate